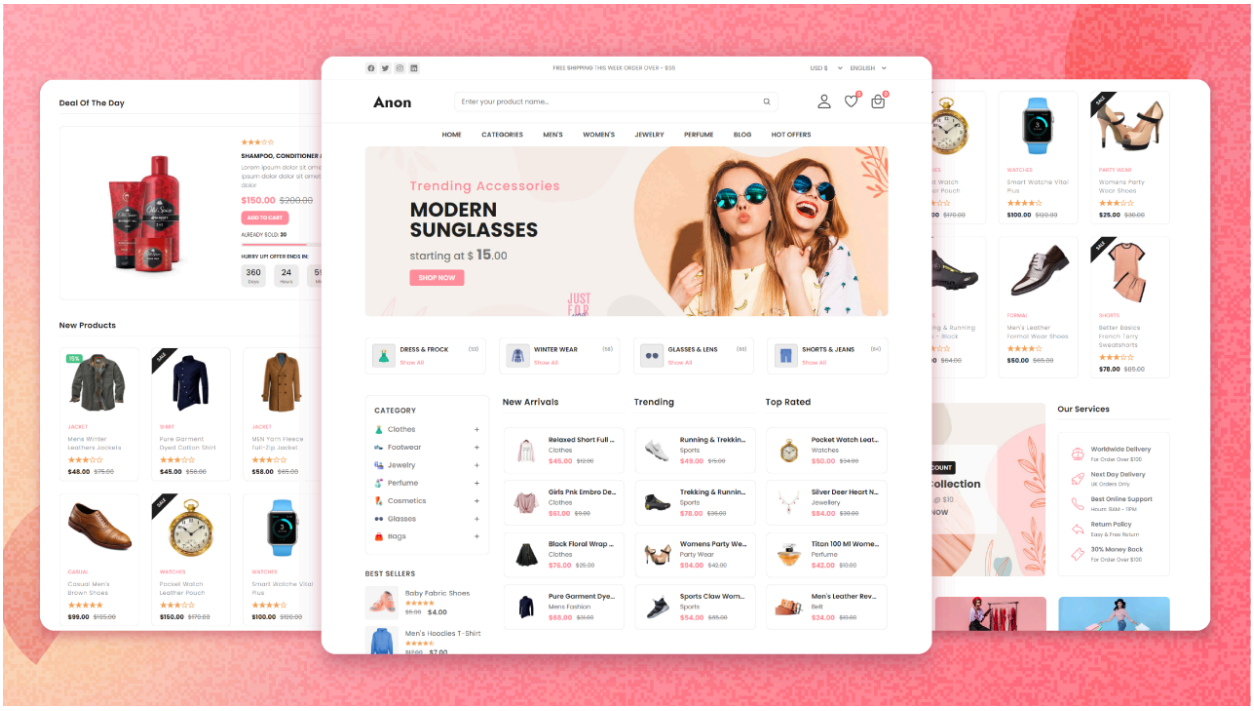
Project 1: Build E-commerce Website

Project Overview



The E-commerce Website is a comprehensive web application designed to provide users with an online shopping experience. This system allows customers to browse products, add items to the cart, and complete purchases while providing administrators with tools to manage products, users, and monitor sales.

Project Duration

4.5 days ⇒ 110 hour

Technology Stack

including, but not limited to:

- PHP (Object-Oriented)
- MySQL
- HTML5
- CSS3 (Bootstrap)
- Git (Version Control)

Business Requirements

Create a full-featured E-commerce Website that enables users to shop online and administrators to manage products, categories, users, and monitor sales through a comprehensive dashboard.

Core Features

User Management

including, but not limited to:

- User registration and authentication system
- User profile management
- Role-based access control (Admin and Regular User roles)

Product Management

- Product browsing and detailed information
- Product search and filtering by name, category, price
- Product comments and ratings

Shopping System

including, but not limited to:

- Add to Cart, View Cart
- Update or Delete Items from Cart
- Checkout Process (Complete order)

Administrative Features

- Comprehensive admin dashboard with a set of insights including, but not limited to:
  - Total Income
  - Number of Categories
  - Number of Orders
  - Number of Customers
  - Daily Revenue Table
- Full CRUD Operations:
  - Products
  - Categories
  - Users
  - Orders
- Product reviews and ratings management

Technical Requirements

System Architecture

- OOP architecture is mandatory! (MVC optional but encouraged)
- Responsive design for multiple devices
- Session-based authentication
- Database-driven content management (Using PDO)

Database Design

- Implement a relational MySQL database with appropriate tables
- Follow proper database design principles
- Implement a secure data access layer

Security Requirements

- Password security (Encryption)
- Input validation
- Secure code using prepared statements
- Session management
- Secured APIs (Protected Routes)
- Immune to SQL Injection and XSS Attacks

UI/UX Design Requirements

- Create wireframes and mockups for key pages
- Implement responsive design (mobile-friendly)
- Apply UX/UI principles
- Landing page with:
  - Navbar (Login/Register)
  - Slider for promotions
  - Discount Section

Implementation Phases

Phase 1: Setup and Planning

- Set up development environment (XAMPP, VS Code, MySQL, Git)
- Create database schema
- Design the Database ERD

- Design system architecture
- Create wireframes for key pages

Phase 2: Core Functionality

- Implement a user authentication system
- Develop a basic admin dashboard
- Create product management functionality
- Implement category management
- AI Integrated Feature (example: an AI Shopping assistant where you explain what type of clothes you are looking for and for what use, and it will return/suggest one or more items that match what you look for) ⇒ **input example** ⇒ I am 182cm tall, and I weigh 105kg. I need a new training suit for sport and working out, and I prefer dark-colored clothes.
- Google Auth for register/login besides the regular email/password form

Phase 3: Shopping Features

- Develop product listing and details pages
- Implement shopping cart functionality
- Create checkout process
- Develop order management system

Phase 4: Testing

- Conduct comprehensive testing
- Fix identified bugs and issues
- Optimize performance

Deliverables

- Wireframe and mock-up files
- Trello Board Link with task breakdown for each team member

- Live link or GitHub Repository link
- Presentation slide deck for the Group Presentation.
- The Database ERD

Evaluation Criteria

As specified in the assessment criteria table, apply Rubric Grading:

- Create mock-ups for an application (Level 3)
- Create static/adaptive UI (Level 3)
- Develop dynamic UI (Level 3)
- Create a database (Level 1)
- Develop data access layer (Level 1)
- Build back-end logic (Level 1)

Learning Outcomes

- Develop proficiency in PHP (OOP) and MySQL
- Gain experience in building secure web applications
- Learn database design and implementation
- Understand user authentication and authorization
- Practice responsive web design principles
- Develop project management skills

Resources

- W3Schools PHP
- How to Create an E-commerce Website in PHP
- Open-source Admin & E-commerce Bootstrap Templates

Important Notes

- Implement proper session management in login & signup
- Follow object-oriented programming principles
- Ensure proper validation for all user inputs
- Implement responsive design for mobile compatibility
- Deadline: Saturday, October 31th, 2026 — 10:00 PM